approach we must also include the “pilot” wave, an object that is also spread throughout a wealth of other locations. (The expert reader should note that the pilot wave is just the wavefunction of conventional quantum mechanics, although its incarnation in Bohm’s theory is rather different.) To avoid endless qualifications, the discussion that follows will be from the perspective of conventional quantum mechanics (the approach most widely used), leaving remarks on Bohm’s and other approaches to the last part of the chapter.

3. For a mathematical but highly pedagogical account see R. P. Feynman and A. R. Hibbs, *Quantum Mechanics and Path Integrals* (Burr Ridge, Ill.: McGraw-Hill Higher Education, 1965).

4. You might be tempted to invoke the discussion of Chapter 3, in which we learned that at light speed time slows to a halt, to argue that from the photon’s perspective all moments are the same moment, so the photon “knows” how the detector switch is set when it passes the beam-splitter. However, these experiments can be carried out with other particle species, such as electrons, that travel slower than light, and the results are unchanged. Thus, this perspective does not illuminate the essential physics.

5. The experimental setup discussed, as well as the actual confirming experimental results, comes from Y. Kim, R. Yu, S. Kulik, Y. Shih, M. Scully, Phys. Rev. Lett, vol. 84, no. 1, pp. 1–5.

6. Quantum mechanics can also be based on an equivalent equation presented in a different form (known as matrix mechanics) by Werner Heisenberg in 1925. For the mathematically inclined reader, Schrödinger’s equation is: $H\,\Psi(x,t) = i\hbar\,(d\Psi(x,t)/\,dt)$, where H stands for the Hamiltonian, Ψ stands for the wavefunction, and $\hbar$ is Planck’s constant.

7. The expert reader will note that I am suppressing one subtle point here. Namely, we would have to take the complex conjugate of the particle’s wavefunction to ensure that it solves the time-reversed version of Schrödinger’s equation. That is, the T operation described in endnote 2 of Chapter 6 takes a wavefunction $\Psi(x,t)$ and maps it to $\Psi^*(x,-t)$. This has no significant impact on the discussion in the text.